Zihao Wang

Date of birth:2002.4.22
Address: Luoyang Town, Suizhou City, Hubei Province, China

+86 -18771363558
wangzihao2020@lzu.edu.cn
Blog:https://www.cnblogs.com/hetailang/

Education

•Computer Science and Technology, Lanzhou University(LZU, Project 985) *2020 - 2024*

- **GPA:** 89
- **Ranking:** top15%
- **IELTS:** 7(8,8,6.5,6)

Awards and Honors

- Lanzhou University Third-Class Scholarship, 2021
- Lanzhou University Second-Class Scholarship, 2022
- Lanzhou University School Level Learning Pacesetter, 2022
- Team Second Prize of the Group Programming Ladder Tournament(GPLT), 2022
- Individual Third Prize of the Group Programming Ladder Tournament(GPLT), 2023
- First Prize of the Lan Qiao Cup Provincial Competition, 2022&2023
- Third Prize of the Lan Qiao Cup National Championship, 2022

Research Experience

•2D Management of KV-Cache in LLM Inference via Layer-wise Optimal Budget *2023.10 - 2024.2*

Internship Project

- During the inference computation process of decoder-only large language models, the inherent data dependencies and the extremely large parameter scale of the algorithms lead to low GPU utilization. This hinders the efficient deployment and inference of large language models in practical applications. Especially when the text length is long, the inference computation overhead increases rapidly at a quadratic rate relative to the text length, further exacerbating the system challenges for long-text applications. Based on the current efficient inference algorithms for large language models, such as model pruning, KV sparsification, and batch task optimization, this project aims to explore how to further optimize the inference performance of large language models by deeply analyzing the numerical characteristics of the multi-layer transformer architecture, thereby alleviating the performance bottlenecks caused by GPU I/O at various levels
- The paper has been submitted to NeurIPS 2024
- Arxiv link: SqueezetionAttention

•Optimization and Scheduling of UAV Energy Efficiency for Grassland Restoration *2022.4 - 2023.4*

National College Student Innovation and Entrepreneurship Training Program Project

- We build a model, which can be seen as the combination of TSP problem and Knapsack problem, and propose a heuristic algorithm for solving
- I am mainly responsible for code writing and testing. The project has been closed

•Research Training *2021.9 - 2024.6*

Team Meeting

- Joined Dongbin Jiao's research group of Lanzhou University
- During this period, I conducted systematic research training, read a certain amount of English literature, reproduced some experimental codes, and conducted tests. I participated in weekly group meetings and reported on the current research progress at the meetings

Skills

Languages: C, C++, Python, SQL
Language Proficiency: Chinese (native level), English (IELTS:7(8,8,6.5,6))
Adobe Pagemaker: LaTeX, Word

Extracurricular Activities

•**Lanzhou University ACM Training,** ACM Team Member *2021 - 2022*

- Participated in Lanzhou University ACM Competition Training And acquired a lot of algorithms, programming skills
- Gained experience in analyzing and solving complex programming challenges within time constraints
- Strengthened critical thinking and troubleshooting skills while working on a wide range of problem domains
- Acquired a deep understanding of efficient coding practices and optimization techniques

•**Lanzhou University Summer school,** Teaching Assistant *2022.7*

- Assist the teacher in guiding the basic programming knowledge of freshmen
- Explain basic programming language (C/C++) to students
- Assign programming assignments to students and correct error codes